CLINIQUE SYSTEM TEST MATRIX:

This document outlines the functional and system data testing scenarios for the CliniQueue system to evaluate its overall functionality, reliability, performance, and compliance with the defined system requirements.

AUTHENTICATION & ROLE AUTHENTICATION

Test ID	Testing level	Test scenario	Test data	Expected Result	Status
AUTH-01	**Integration**	Admin sends a registration link to a valid DLSU-D account	Valid DLSU-D email	Registration link is sent to the intended email	**Success | Fail | Pending**
AUTH-02	**E2W**	Admin attempts to send a registration link to a non-DLSU-D email	Personal email	Registration is rejected	**Success | Fail | Pending**
AUTH-03	**E2E**	User opens registration link	Registration link & form	Registration page opens successfully	**Success | Fail | Pending**
AUTH-04	**E2E**	User registers with valid information	DLSU-D email, password, and required fields	Account is created successfully	**Success | Fail | Pending**
AUTH-05	**Unit**	User attempts registration with an expired link	Expired registration link	Registration is denied; user must request a new link	**Success | Fail | Pending**
AUTH-06	**Unit**	User attempts to reuse a registration link	Link has already been used	Duplicate registration is prevented	**Success | Fail | Pending**
AUTH-07	**Unit**	User submits incomplete registration details	Missing required fields	Validation errors are displayed	**Success | Fail | Pending**
AUTH-08	**E2E**	User submits a weak password	Password that does not meet	Validation errors are displayed	**Success | Fail | Pending**

			requirements		
AUTH-09	**E2E**	User signs in with valid credentials	Registered account and correct password	User is authenticated and redirected to the appropriate dashboard	**Success | Fail | Pending**
AUTH-10	**E2E**	User attempts to sign in before completing registration	User	Access is denied	**Success | Fail | Pending**
AUTH-11	**E2E**	User attempts to access a protected page without signing in	No active authenticated session	User is redirected to sign-in or denied access	**Success | Fail | Pending**
AUTH-12	**E2E**	User attempts to access another role's restricted page	Authenticated user without the required role	Access is denied	**Success | Fail | Pending**
AUTH-13	**E2E**	User signs out and revisits a protected page	Previously authenticated user who has signed out	Protected content is no longer accessible without signing in again	**Success | Fail | Pending**
AUTH-14	**Integration**	User requests a password reset using a registered email	Registered email	Password reset instructions are sent to the registered email	**Success | Fail | Pending**
AUTH-15	**E2E**	User opens a valid password reset link	Valid, unused reset link	Password reset page opens successfully	**Success | Fail | Pending**
AUTH-16	**Unit**	User opens an expired	Expired reset link	Password reset is	**Success | Fail |**

		password reset link		denied; user is prompted to request a new link	**Pending**
AUTH-17	**Unit**	User reuses a password reset link	Previously used reset link	Password reset is denied	
AUTH-18	**E2E**	User resets password with valid input	Valid reset link and password meeting requirements	Password is updated successfully	**Success | Fail | Pending**
AUTH-19	**Unit**	User submits a weak or invalid new password	Password that does not meet requirements	Validation error is displayed and password remains unchanged	**Success | Fail | Pending**
AUTH-20	**Unit**	User submits mismatched new passwords	Password and confirmation fields differ	Validation error is displayed and password is not changed	**Success | Fail | Pending**
AUTH-21	**E2E**	User signs in using the new password	New password	Login succeeds with the new password	**Success | Fail | Pending**
AUTH-22	**E2E**	User signs in using the old password	Old password	Login is rejected using the old password	**Success | Fail | Pending**
AUTH-23	**Integration**	User requests multiple password reset links	Several reset requests are submitted for the same account	Reset tokens follow the system's defined invalidation and expiration policy; only valid tokens can be used	**Success | Fail | Pending**

APPOINTMENT & BOOKING MANAGEMENT

Test ID	Testing level	Test scenario	Test data	Expected Result	Status
BOOK-01	**E2E**	Patient books an available appointment slot	Valid patient account, available date and time slot, complete booking details	Appointment is created successfully and a booking confirmation is displayed	**Success | Fail | Pending**
BOOK-02	**Unit**	System validates required booking fields	Missing patient information, appointment date, or required booking details	Validation errors are displayed and invalid booking data is rejected	**Success | Fail | Pending**
BOOK-03	**Integration**	System updates slot availability after a successful booking	System updates slot availability after a successful booking	Slot availability is updated in the database and reflected in the booking interface	**Success | Fail | Pending**
BOOK-04	**Unit&Integration**	Patient attempts to book a fully booked slot	Appointment slot with no remaining capacity	Booking is rejected and the system informs the patient that the slot is unavailable	**Success | Fail | Pending**
BOOK-05	**Integration**	Two patients attempt to book the final available slot simultaneously	Two patient accounts and one remaining slot	Only one booking succeeds, and the slot is not overbooked	**Success | Fail | Pending**
BOOK-06	**Unit**	System validates appointment date and time	Past date, invalid time, or date outside the permitted booking period	Invalid appointment schedules are rejected according to the configured scheduling	**Success | Fail | Pending**

				rules	
BOOK-07	**E2E**	Patient views their existing appointments	Authenticated patient with one or more existing bookings	Only the authenticated patient's appointments are displayed	**Success | Fail | Pending**
BOOK-08	**E2E**	Patient cancels an existing appointment	Authenticated patient with an active appointment	Appointment status is updated to cancelled and the change is reflected in the booking records	**Success | Fail | Pending**
BOOK-9	**Integration**	System releases a slot after appointment cancellation	Active booking occupying a slot with available capacity after cancellation	The cancelled booking no longer occupies the slot, making it available according to scheduling rules	**Success | Fail | Pending**
BOOK-10	**E2E**	Patient reschedules an existing appointment	Active booking and a different available appointment slot	The original booking is updated or replaced according to the system's rescheduling rules, and the new schedule is recorded	**Success | Fail | Pending**
BOOK-11	**E2E**	Patient attempts to reschedule to an unavailable slot	Active booking and a fully booked destination slot	Rescheduling is rejected and the original appointment remains unchanged	**Success | Fail | Pending**

BOOK-12	**Unit & Integration**	System prevents duplicate appointment submissions	Same patient, same appointment details, and repeated submission of the same request	Duplicate submissions do not create unintended duplicate bookings	**Success | Fail | Pending**
BOOK-13	**Unit**	System validates booking capacity	Slot capacity, current confirmed bookings, and a new booking request	The system accepts bookings within capacity and rejects bookings that exceed it	**Success | Fail | Pending**
BOOK-14	**E2E**	Walk-in patient requests an appointment at the clinic	Patient details, requested clinic or service, and an available appointment slot	Walk-in patients are successfully registered and added to the active queue alongside patients with online bookings.	**Success | Fail | Pending**
BOOK-15	**E2E**	Clinic staff views scheduled appointments	Authenticated clinic staff account and existing appointment records	Clinic staff can view appointments within their assigned clinic and authorized scope	**Success | Fail | Pending**

QUEUE MANAGEMENT:

Test ID	Testing Level	Test Scenario	Test data	Expected result	Status
QUEUE-01	**Unit**	Generate a queue number for a patient	Patient with a confirmed appointment or a registered walk-in entry	A unique queue number is generated for the patient.	**Success | Fail | Pending**
QUEUE-02	**Integration**	Add a	Valid	The patient	**Success |**

		patient to the queue after a successful booking	appointment and available queue slot	is added to the correct queue and associated with the appointment.	**Fail | Pending**
QUEUE-03	**E2E**	Patient joins the queue successfully	Registered patient with a valid appointment or eligibility to queue	The patient receives a queue number and confirmation of successful entry.	**Success | Fail | Pending**
QUEUE-04	**Unit**	Maintain the correct queue order	Multiple patients with different queue positions or priorities	Patients are arranged according to the system's defined queue rules.	**Success | Fail | Pending**
QUEUE-05	**Integration**	Update a patient's queue status	Patient currently marked as Waiting	The status is updated correctly and reflected in the queue display.	**Success | Fail | Pending**
QUEUE-06	**E2E**	Staff notify the next patient	Multiple patients are waiting in the queue	The next eligible patient is identified and their queue status changes to Called.	**Success | Fail | Pending**
QUEUE-07	**Integration**	Prevent duplicate active queue entries	Patient already has an active queue entry for the same appointment or service	The system prevents an unintended duplicate entry.	**Success | Fail | Pending**
QUEUE-08	**Unit**	Handle an empty queue	No patients are waiting	The system displays an appropriate empty-queue message and does not attempt to call a	**Success | Fail | Pending**

				nonexistent patient.	
QUEUE-09	**E2E**	Mark a patient as served	Patient has been called and consultation is completed	The patient's status changes to Served, and the queue advances to the next eligible patient.	**Success | Fail | Pending**
QUEUE-10	**Integration**	Handle a skipped or absent patient	Patient is notified but does not respond	Staff can mark the patient as Skipped or No-show, and the queue proceeds according to the defined rules.	**Success | Fail | Pending**
QUEUE-11	**Integration**	**Synchronize queue updates across user interfaces**	**Staff updates a patient's queue status while the patient dashboard is open**	**Relevant displays show the updated queue information without inconsistent status values.**	**Success | Fail | Pending**

NOTIFICATION MANAGEMENT

Test ID	Testing Level	Test Scenario	Test Data	Expected Result	Status
NOTIF-01	**E2E**	Patient receives an appointment booking confirmation	Valid patient account, available appointment slot, and complete booking details	Appointment is booked successfully, and a confirmation notification is displayed or sent through the configured notification channel.	**Success | Fail | Pending**

NOTIF-02	**Integration**	Patient receives an appointment cancellation notification	Patient with an existing appointment and a successfully processed cancellation	The system sends or displays a cancellation notification containing the correct appointment details.	**Success | Fail | Pending**
NOTIF-03	**Integration**	Patient receives an appointment rescheduling notification	Existing appointment and a successfully rescheduled appointment	The system notifies the patient of the updated appointment date and time.	**Success | Fail | Pending**
NOTIF-04	**Integration**	Patient receives a notification when their queue status changes	Patient with an active queue entry and a queue status update	The system sends or displays a notification reflecting the patient's latest queue status.	**Success | Fail | Pending**
NOTIF-05	**E2E**	Patient receives a notification when their queue number is called	Patient with a queue entry marked as Waiting and clinic staff authorized to call the patient	The patient's queue number is called, and the correspondin g notification is delivered or displayed through the configured channel.	**Success | Fail | Pending**
NOTIF-06	**Integration**	System prevents duplicate notifications for the same event	One appointment booking or queue update that triggers a notification	The system does not send unintended duplicate notifications for the same event.	**Success | Fail | Pending**
NOTIF-07	**Integration**	System handles notification delivery failure	Valid notification event and an unavailable email, SMS,	The system handles the delivery failure appropriately	**Success | Fail | Pending**

			or notification service	, records the failure when supported, and retries or reports it according to the configured rules.	
NOTIF-08	**Unit**	System validates notification content	Notification template with patient details, appointment information, or queue number	The generated notification contains the correct values and required information without missing or incorrectly substituted fields.	**Success | Fail | Pending**
NOTIF-09	**Integration**	Clinic staff receives relevant appointment or queue notifications	Authenticated clinic staff account and an event within the staff member's authorized clinic scope	The system delivers or displays the relevant notification to the appropriate staff member without exposing unrelated patient information.	**Success | Fail | Pending**
NOTIF-10	**E2E**	Patient views their notifications	Authenticated patient account with existing notifications	The patient can view their own notifications, and the displayed information corresponds to their appointments or queue activity.	**Success | Fail | Pending**
NOTIF-11	**Unit**	System marks a notification	Existing unread notification	The notification's read status	**Success | Fail | Pending**

		as read	belonging to the authenticated user	changes correctly without modifying unrelated notifications.	

DATA INTEGRITY & DATABASE MANAGEMENT

Test ID	Testing Level	Test Scenario	Test Data	Expected Result	Status
DATA-01	**Integration**	System saves patient and appointment records to the database	Valid patient information and complete appointment booking details	Patient and appointment records are saved correctly and can be retrieved from the database.	**Success | Fail | Pending**
DATA-02	**Integration**	System updates database records after appointment cancellation	Existing appointment with an active status	The appointment status is updated to Cancelled, and the change is reflected in the database.	**Success | Fail | Pending**
DATA-03	**Integration**	System updates database records after appointment rescheduling	Existing appointment and a valid new appointment schedule	The updated appointment details are saved correctly without leaving unintended duplicate or conflicting active bookings.	**Success | Fail | Pending**
DATA-04	**Integration**	System maintains consistency between appointments and queue entries	Patient with a confirmed appointment or registered walk-in entry	The patient's queue entry is correctly associated with the corresponding patient and	**Success | Fail | Pending**

				appointment or walk-in record.	
DATA-05	**Unit**	System validates data before saving records	Missing required fields, invalid values, or improperly formatted data	Invalid data is rejected before being accepted by the relevant application logic.	**Success | Fail | Pending**
DATA-06	**Integration**	System prevents duplicate patient records	Existing patient records	The system prevents unintended duplication of patient records	**Success | Fail | Pending**
DATA-07	**Integration**	System preserves existing data when an operation fails	Existing database record and a simulated failure during an update or transaction	The system avoids partially applied changes and preserves data consistency according to the transaction rules.	**Success | Fail | Pending**
DATA-08	**Integration**	System handles database connection failure	Simulated unavailable database connection during a read or write operation	The system handles the failure gracefully, displays an appropriate error, and does not falsely report that the operation succeeded.	**Success | Fail | Pending**
DATA-09	**Integration**	System retrieves the latest saved data	Existing record that has been successfully created or updated	The system retrieves the latest saved values rather than displaying outdated or inconsistent information.	**Success | Fail | Pending**

DATA-10	**Integration**	System prevents unauthorized data modifications	Patient account attempting to modify protected patient, appointment, or queue records	Unauthorized modifications are rejected, and the original data remains unchanged.	**Success | Fail | Pending**
DATA-11	**Integration**	System maintains data consistency during simultaneous updates	Two authorized requests attempting to modify the same appointment or queue record	The system applies its defined concurrency rules and prevents conflicting or inconsistent final records.	**Success | Fail | Pending**
DATA-12	**Integration**	System preserves existing records when related records are cancelled or deactivated	Patient with existing appointments or queue history	Related records remain consistent and historical information is retained according to the system's data-retention rules.	**Success | Fail | Pending**

USER & ADMINISTRATIVE MANAGEMENT

Test ID	Testing Level	Test Scenario	Test Data	Expected Result	Status
ADMIN-01	**E2E**	Administrator views registered user accounts	Authenticated administrator account and existing user records	The administrator can view the list of registered users and their relevant account information.	**Success | Fail | Pending**
ADMIN-02	**Integration**	Administrator assigns or updates a user's role	Existing user account and a valid role assignment	The user's role is updated and saved correctly,	**Success | Fail | Pending**

				and the assigned permissions reflect the new role.	
ADMIN-03	**E2E**	Administrator deactivates a user account	Existing active user account and an authorized administrator	The account is deactivated successfully, and the user can no longer access protected features requiring an active account.	**Success | Fail | Pending**
ADMIN-04	**E2E**	Administrator reactivates a deactivated user account	Existing deactivated user account and an authorized administrator	The account is reactivated, and access is restored according to the assigned role and account rules.	**Success | Fail | Pending**
ADMIN-05	**Integration**	System prevents unauthorized users from performing administrative actions	Authenticated patient or clinic staff account attempting an administrator-only action	The action is rejected, and administrative records remain unchanged.	**Success | Fail | Pending**
ADMIN-06	**Integration**	Administrator updates clinic configuration settings	Authorized administrator and valid configuration values	The updated settings are saved and applied to the relevant system functions.	**Success | Fail | Pending**
ADMIN-07	**Unit**	System validates administrative configuration inputs	Missing, invalid, or out-of-range configuration values	Invalid configuration values are rejected, and validation feedback is	**Success | Fail | Pending**

				provided.	
ADMIN-08	**E2E**	Administrator searches for a specific user account	Existing user account and a supported search term, such as email	The system displays the matching account or an appropriate message when no account matches.	**Success | Fail | Pending**

DASHBOARD & REPORTING MANAGEMENT

Test ID	Testing Level	Test Scenario	Test Data	Expected Result	Status
DASH-01	**E2E**	Clinic staff views the dashboard	Authenticated clinic staff account and existing clinic records	The dashboard loads successfully and displays information available to the staff member's authorized scope.	**Success | Fail | Pending**
DASH-02	**Integration**	Dashboard displays the correct appointment count	Known set of appointments with different statuses	The displayed appointment count matches the underlying records and the dashboard's counting rules.	**Success | Fail | Pending**
DASH-03	**Integration**	Dashboard displays the correct active queue count	Known set of queue entries with different statuses	The displayed active queue count matches the number of entries considered active under the system's rules.	**Success | Fail | Pending**

DASH-04	**Integration**	Dashboard reflects appointment updates	An appointment is booked, cancelled, or rescheduled	The dashboard reflects the relevant change according to its refresh and update behavior.	**Success | Fail | Pending**
DASH-05	**Integration**	Dashboard reflects queue status updates	Queue entries in Waiting, Called, In Consultation, or Served states	Queue summaries and counts reflect the current status of queue entries.	**Success | Fail | Pending**
DASH-06	**E2E**	Administrator views clinic activity statistics	Authorized administrator account and records covering a defined reporting period	The dashboard displays clinic activity statistics for the selected period and authorized scope.	**Success | Fail | Pending**
DASH-07	**Integration**	System filters dashboard data by date range	Existing appointment and queue records across different dates	Only records matching the selected date range are included in the relevant summaries.	**Success | Fail | Pending**
DASH-08	**Integration**	System generates an appointment or queue report	Existing records and a supported report type	The generated report contains accurate records and totals for the selected report criteria.	**Success | Fail | Pending**
DASH-09	**E2E**	Authorized user exports a report	Authorized account, available report, and supported	The report is exported successfully in the supported	**Success | Fail | Pending**

			export format	format, and its contents match the selected report data.	
DASH-10	**Integration**	System restricts dashboard data according to user role	Patient, clinic staff, and administrator accounts with different permissions	Each user can access only the dashboard information permitted by their assigned role.	**Success | Fail | Pending**